

Linear programming

Problems which seek to maximize or minimize a numerical function of number of variables subject to certain conditions or constraints, form a general class called optimization problems. Of all operation research model, linear programming is one with more versatile, powerful and useful technique. It is the general technique for optimum allocation of scarce or limited resources (eg: labor, material, machine, capital and energy) to several competing activities (eg: production services, jobs, new equipment and projects) on the basis of given criteria of optimality.

The word linear is used to describe the proportionate relationship of two or more variables in model. Thus a given change in one variable will always cause a resulting proportional change in another variable i.e there exists a linear relationship among several activities (or variables). For example, doubling the investment on certain project, will exactly double the rate of return.

The word programming here is used to specify a sort of planning that involves the economic allocation of limited resources by adopting a particular course of action or strategy amongst various alternatives /strategies to achieve the desired objective.

The general mathematical model of LPP

The general linear programming problem with n decision variables and m constraints can be stated in following form:

To find values of decision variables $x_1, x_2, \dots, x_n$ such that optimize(Maximize/minimize)

$$Z = c_1x_1 + c_2x_2 + \dots + c_nx_n \quad \text{subject to linear constraints}$$

$$a_{11}x_1 + a_{12}x_2 + \dots + a_{1n}x_n (\leq, =, \geq) b_1$$

$$a_{21}x_1 + a_{22}x_2 + \dots + a_{2n}x_n (\leq, =, \geq) b_2$$

.....

.....

$$a_{m1}x_1 + a_{m2}x_2 + \dots + a_{mn}x_n (\leq, =, \geq) b_m$$

$$\text{And } x_1 \geq 0, x_2 \geq 0, \dots, x_n \geq 0.$$

The above formulation can also be expressed in compact form using summation sign

$$\text{Optimize (Max/Min) } Z = \sum_{i=1}^n c_i x_i \quad (\text{objective function})$$

$$\text{Subject to constraints } \sum_{j=1}^n a_{ij} x_j (\leq, =, \geq) b_i, \quad i = 1, 2, \dots, m$$

$$\text{And } x_j \geq 0, \quad j = 1, 2, \dots, n \quad (\text{non-negativity constraint})$$

Remark : In the general problem, the expression ($\leq, =, \geq$) means that in any specific problem each constraint may take any one of three possibilities.

Here, c_j 's are coefficients representing the per unit contribution of decision variable x_j to value of objective function.

The a_{ij} 's are called input-output coefficients. These represent the amount of resources that is consumed per unit of variable x_j . In given constraint a_{ij} 's can be positive, negative or zero. The

b_i 's represent the total availability of i^{th} resource. The term resource is used in a very general sense to include any numerical value associated with RHS of a constraint. It is assumed that $b_i \geq 0$ for all i .

However, if any $b_i < 0$, then both sides of constraint i can be multiplied by -1 to make $b_i > 0$ and reverse the inequality of that constraint.

Any maximization LPP can be converted into an equivalent minimization LP problem and vice-versa by multiplying the given objective function by -1 without making any change in the constraint

Example: $\min Z = \sum_{i=1}^n c_i x_i$ is equivalent to $\max Z^* = \sum_{i=1}^n (-c_i) x_i$ where $Z = -Z^*$

Slack and surplus variables

In any LPP, the constraints may take any one of three possibilities ($\leq, =, \geq$). However, since in general, it is easy to work with equalities than inequalities, the constraints of LPP which are expressed as system of linear inequalities may be converted into equalities by adding arbitrary additional non-negative variables, called slack and surplus variables.

If constraint is of the form $\sum_{j=1}^n a_{ij} x_j \leq b_i$ then it can be converted to

$\sum_{j=1}^n a_{ij} x_j + s_i = b_i, i = 1 \text{ to } m$, here s_i , a non-negative variable is termed as slack variable.

If constraint is of the form $\sum_{j=1}^n a_{ij} x_j \geq b_i$ then it can be converted to

$\sum_{j=1}^n a_{ij} x_j - s_i = b_i, i = 1 \text{ to } m$, here s_i , a non-negative variable is termed as surplus variable.

Remark : The coefficient of slack and surplus variables are zero in objective function due to the reason that they represent unused capacity or resource.

Formulation of LP models

Problems

1. A company makes 3 products x, y and z which pass through 3 departments drill, lathe and assembly. The number of hours available in each department and number of hours required by each product in each department and profit contribution of each product are given as following.

	Time required in hrs			Profit (Rs)
	Drill	Lathe	Assembly	
X	3	3	8	9
Y	6	5	10	15
Z	7	4	12	20

Hours available	210	240	260	
-----------------	-----	-----	-----	--

Formulate the above as a LPP

Solution:

Let x_1, x_2, x_3 denote number of units of product x, y, z respectively.

The objective is to maximize profit. Hence, the objective function can be expressed as

Maximize $Z = 9x_1 + 15x_2 + 20x_3$ subject to the following constraints

a) Constraint on number of hours available in drilling department

$$3x_1 + 6x_2 + 7x_3 \leq 210$$

b) Constraint on number of hours available in Lathe department

$$3x_1 + 5x_2 + 4x_3 \leq 240$$

c) Constraint on number of hours available in Assembly department

$$8x_1 + 10x_2 + 12x_3 \leq 260$$

Finally, all the units produced must be nonnegative

$$\text{Hence, } x_1 \geq 0, x_2 \geq 0, x_3 \geq 0$$

2. A farmer has 100 acres of farm. He can sell all tomatoes, potatoes and beans and he can raise the price to obtain Rs 1 per kg of tomatoes, Rs. 0.75 per kg of potatoes and Rs 2 per kg for beans. The average yield per acre is 2000 kg of tomatoes, 3000 kg of potatoes and 1000 kg of beans. Fertilizers are available at Rs 0.50 per kg and amount required per acre is 100 kgs to each tomatoes and potatoes and about 50 kgs for beans. Labor required for cultivating and harvesting per acre is 5 man /day for tomato and beans. Also 6 man/day for potatoes. A total of 400 men/ day are available at Rs 20 per day. Formulate this problem as LP model to maximize farmer's total profit.

Solution:

Let x_1, x_2, x_3 denote number of acres of farmland used to grow tomatoes, potatoes and beans respectively.

The total sales is : average yield per acre x price per kg x Number of acres

$$1(2000)x_1 + 0.75(3000)x_2 + 2(1000)x_3$$

Expenditure on fertilizer and labor

$$\text{Fertilizer: } (100x_1 + 100x_2 + 50x_3)0.5$$

$$\text{Labor: } (5x_1 + 6x_2 + 5x_3)20$$

Thus , the profit function is $Z = \text{Total sales} - \text{Total expenditure}$

$$Z = (2000x_1 + 2250x_2 + 3000x_3) - (100x_1 + 120x_2 + 100x_3) - (50x_1 + 50x_2 + 25x_3)$$

$$\text{Maximize } Z = 1850x_1 + 2080x_2 + 1875x_3$$

Subject to the constraints a) limited acres of land $x_1 + x_2 + x_3 \leq 100$

b) limited laborers $5x_1 + 6x_2 + 5x_3 \leq 400$

c) non-negativity condition $x_1 \geq 0, x_2 \geq 0, x_3 \geq 0$

3. The owner of the metro sports wishes to determine how many advertisement to place in three magazines A, B, C. His objective is to advertise in such a way that the total exposure to principle buyers of expensive sports goods is to maximize percentage of readers per each magazine is known. Exposure in any magazine is the number of advertisements placed

multiply by the principle buyers.

	Magazines		
	A	B	C
Readers	1 lakh	0.6 lakh	0.4lakh
Principle Buyers	10%	15%	7%
Cost per advertisement	5000	4500	4250

The budgeted amount is at most Rs 1 lakh for advertisement. The owner has already decided that magazine A should have not more than 6 advertisement and each B and C magazines have at least 2 advertisement. Formulate LPP.

Solution:

Let x_1, x_2, x_3 denote total number of advertisements placed in magazines A, B, C respectively.

Maximize $Z = 1,00,000 (.10)x_1 + 60,000(.15)x_2 + 40,000(0.07)x_3$

$Z = 10,000x_1 + 9000x_2 + 2800x_3$ subject to the constraints

$5,000x_1 + 4500x_2 + 4250x_3 \leq 1,00,000$ (budgeted amount available is 1 lakh)

$x_1 \leq 6$ (magazine A should not have more than 6 ads)

$x_2 \geq 2$ (Magazine B and C have at least 2 ads)

$x_3 \geq 2$

Also $x_1 \geq 0, x_2 \geq 0, x_3 \geq 0$

4. A paper mill produces rolls of paper used in making cash registers. Each roll of paper is 100 meter in length and can be used in width of 3, 4, 6 and 10 centimeters. The company production process results in rolls that are 24cm in width. The company must cut its 24cm roll to the desired width. It has 6 basic cutting alternatives as follows.

Cutting alternative	Width of rolls				waste
	3	4	5	6	
1	4	3	-	-	-
2	-	3	2	-	-
3	1	1	1	1	1
4	-	-	2	1	2
5	-	4	1	-	2
6	3	2	1	-	1

Minimum demand for rolls is as follows

Roll width: 3 4 6 10

Demand : 2000 3600 1600 500

The paper mill wishes to minimize the waste resulting from trimming to size. Formulate LP model.

Solution: Let $x_1, x_2, x_3, x_4, x_5, x_6$ be the number of rolls cut by alternatives 1, 2, 3, 4, 5, 6 respectively.

Minimize $Z = 0x_1 + 0x_2 + x_3 + 2x_4 + 2x_5 + x_6$, subject to the demand constraints

$4x_1 + x_3 + 3x_6 \geq 2000$ (for width 3)

$3x_1 + 3x_2 + x_3 + 4x_5 + 2x_6 \geq 3600$ (for width 4)

$2x_2 + x_3 + 2x_4 + x_5 + x_6 \geq 1600$ (for width 6)

$x_3 + x_4 \geq 500$ (for width 10)

And non negative condition $x_1, x_2, x_3, x_4, x_5, x_6 \geq 0$

Graphical solution for LPP with two decision variables

Step 1: State the problem in mathematical form

Step2: By treating the inequalities as equations, plot each equation on a graph. Decide the area of feasible solution according to the inequality sign of constraints. Indicate the area of feasible solution by shaded area.

Step 3: Determine the coordinates of all points at corner of feasible region.

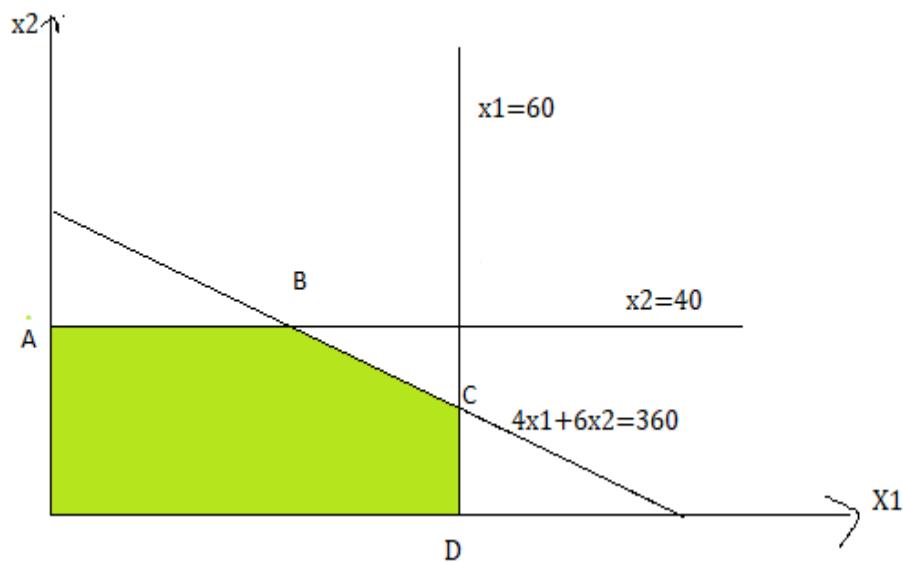
Step 4: Find value of objective function corresponding to solution points determined by step 3.

Step 5: Determine feasible solution which optimize the value of objective function.

Problems.

1) Use graphical method to solve LPP

Maximize $z = 15x_1 + 10x_2$ subject to $4x_1 + 6x_2 \leq 360$, $3x_1 \leq 180$, $5x_2 \leq 200$
 $x_1, x_2 \geq 0$



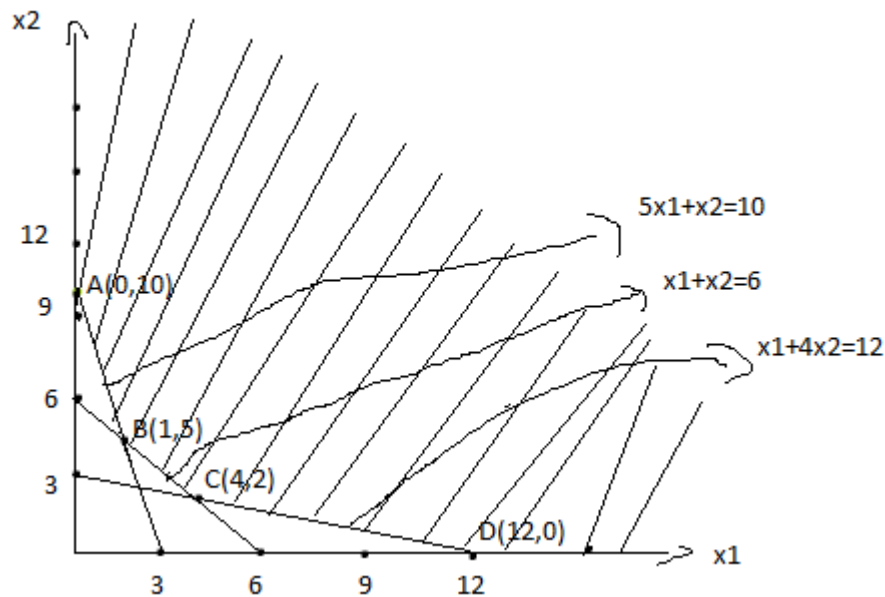
Corner points	$Z = 15x_1 + 10x_2$
A(0,40)	400
B(30,40)	850
C(60, 20)	1100
D(60,0)	900

Observe that the optimal solution of LPP is $x_1=60$, $x_2=20$ and $Z_{\max}=1100$

2. Minimize $z = 3x_1 + 2x_2$ subject to $5x_1 + x_2 \geq 10$; $x_1 + x_2 \geq 6$;

$x_1 + 4x_2 \geq 12$; $x_1, x_2 \geq 0$

Plot the equations $5x_1 + x_2 = 10$; $x_1 + x_2 = 6$; $x_1 + 4x_2 = 12$ on a graph



Corner points	$Z = 3x_1 + 2x_2$
A(0,10)	20
B(1,5)	13
C(4, 2)	16
D(12,0)	36

The optimal solution is $x_1=1$ and $x_2 = 5$. $Z_{\min}=13$

Practice problems:

1. A manufacturer produce 2 types of models M1 and M2. Each M1 model requires four hours of grinding and 2 hours of polishing, whereas each M2 model requires 2 hours of grinding and 5 hours of polishing. The manufacturer has 2 grinders and 3 polishers. Each grinder works for 40 hours per week and each polisher works for 60 hours a week. Profit on M1 model is Rs 3 and on M2 is Rs. 4. Whatever is produced in a week is sold in the market. How Should the manufacturer allocate the production capacity to the two types of models so that he may make the maximum profit in a week.
2. An aero plane can carry a maximum of 200 passengers. A profit of Rs. 400 is made on each first class ticket and a profit of Rs. 300 is made on each economy class ticket. The airline reserves at least 20 seats for first class. However, at least four times as many passengers prefer to travel by economy class than by the first class. How many tickets of each class must be sold in order to maximize profit for the airline? Formulate the problem as an LP model.
3. A company produces two products A and B. The sales volume for A is at least 80% of the total sales of both A and B. However, the company cannot sell more than 110 units of A per day. Both products use one raw material, of which the maximum daily availability is 240 lb. The usage rates of the raw material are 2 lb per unit of A and 4 lb per unit of B. The profit units for A and B are \$10 and \$25, respectively. Determine the optimal product mix for the company.
4. An individual wishes to invest \$5000 over the next year in two types of investment. Investment A yields 5% and investment B yields 8%. Market research recommends an

allocation of at least 25% in A and at most 50% in B. Moreover, investment in A should be at least half the investment in B. How should the fund be allocated to the two investments?

Simplex method

The graphical solution cannot be used for a LPP having more than 2 decision variables. The solution of LPP with more than three decision variables can be obtained by an algebraic iterative method called simplex method.

Some important definitions:

1. Solution: A set of variables $x_1, x_2, \dots, x_{n+m}$ is called a solution to LP problem if it satisfies all m constraints.
2. Feasible solution: A set of variables $x_1, x_2, \dots, x_{n+m}$ is called a feasible solution to LP problem if it satisfies the constraints as well as non-negativity restrictions.
3. Basic solution: A solution obtained by setting any n variables among m+n variables equal to zero and solving for remaining m variables are called basic solution. These m variables are called basic variables and the remaining n variables that have been set to zero each are called non-basic variables. The number of basic solutions thus obtained will be at most $\binom{m+n}{m}$.
4. Basic feasible solution: It is a basic solution that also satisfies the non-negativity restrictions. All bfs ≥ 0 .
5. Non-degenerate solution: It is a basic feasible solution(bfs) in which all the m basic variables are greater than zero and the remaining n variables are zero each.
6. Degenerate solution: It is a bfs in which one or more of the m basic variables are equal to zero.
7. Optimal basic feasible solution: It is the bfs that also optimizes the objective function.
8. Unbounded solution: If the value of the objective function can be increased or decreased indefinitely, then the solution is termed unbounded solution.

George Dantzig, American Mathematician developed an algebraic method of solving LPP of m linear constraints on n decision variables which optimize the objective function.

Simplex method

Step1: Express the LPP in standard form

- a) Check whether the objective function is to be maximized or minimized . If $Z = c_1 x_1 + c_2 x_2 + \dots + c_n x_n$ is to be minimized, then convert it into a problem of maximization by writing $\text{Min } Z = \text{Max}(-Z)$.
- b) Check whether all b_i 's are positive. If any of the b_i 's is negative, multiply both sides of that constraint by -1 so as to make its RHS positive.
- c) Convert all inequalities into equations by introducing slack/surplus variables in the constraints giving equations of the form $a_{i1}x_1 + a_{i2}x_2 + \dots + a_{in}x_n + s_i = b_i, i = 1, 2, \dots, m$

Step 2: Find an initial basic feasible solution.

Let us consider $\text{Max } Z = c_1 x_1 + c_2 x_2 + c_3 x_3$ subject to

$$a_{11}x_1 + a_{12}x_2 + a_{13}x_3 + s_1 = b_1$$

$$a_{21}x_1 + a_{22}x_2 + a_{23}x_3 + s_2 = b_2$$

$$a_{31}x_1 + a_{32}x_2 + a_{33}x_3 + s_3 = b_3$$

$$x_1, x_2, x_3, s_1, s_2, s_3 \geq 0$$

If there are m equations involving n unknowns, then assign zero values to any $(n-m)$ variables for finding a solution. For the above LPP in 3 constraints in 6 six variables, set 3 of the variables to zero. Put up the first simplex table as follows

	C_j	c1	c2	c3	0	0	0		
C_B	Basis	x1	x2	x3	s1	s2	s3	B	Min ratio θ
0	s1	a11	a12	a13	1	0	0	b1	
0	s2	a21	a22	a23	0	1	0	b2	
0	s3	a31	a32	a33	0	0	1	b3	
	$Z_j = \sum C_B a_{ij}$	0	0	0	0	0	0	0	
	$C_j = c_j - Z_j$	c1	c2	c3	0	0	0	0	

Step 4: Applying optimality test

Compute $C_j = c_j - Z_j$ where $Z_j = \sum C_B a_{ij}$. We call C_j row as net evaluation row. It indicates the per unit increase in the objective function if the variable heading the column is brought into the solution. If all C_j are negative, then the initial bfs is optimal. If atleast one C_j is positive, then the current bfs is not optimal and proceed to the next step.

Step 5: i) Identify the incoming and outgoing variables from basis.

If there are more than one positive C_j , then the **incoming variable** is the one that heads the column having maximum C_j . This column is called **key column or pivot column**. If more than one variable has the same maximum C_j , any of these variables may be selected arbitrarily as the incoming variable.

Divide the elements under B column by the corresponding elements of key column and choose the row containing the minimum positive ratio θ . This row is called **key row or pivot row**. The element at the intersection of key row and key column is called **key element or pivot element**. The basic variable in the key row is the **outgoing variable** and the non basic variable in the key column is the **entering variable**. If all ratios are ≤ 0 , the incoming variable can be made as large as we please without violating the feasibility condition. Hence the problem has an unbounded solution and no further iteration is required.

ii) Iterate towards an optimal solution

Drop the outgoing variable and introduce incoming variable along with its associated value under C_B column. Convert the **key element to unity** by dividing the key row by the key element. Then make all other elements of the **key column zero** by subtracting proper multiples of key row from the other rows.

Step 6: Go to step 4 and repeat the computational procedure until either an optimal solution is obtained.

For problems please refer PPT.

Practice problems:

1. Maximize $Z = x_1 - x_2 + 3x_3$ subject to $x_1 + x_2 + x_3 \leq 10, 2x_1 - x_3 \leq 2, 2x_1 - 2x_2 + 3x_3 \leq 0, x_1, x_2, x_3 \geq 0$
Ans : $x_1 = 0, x_2 = 6, x_3 = 4, \text{Max } Z = 6$

2. Maximize $Z = 3x_1 + 2x_2 + 5x_3$ subject to $x_1 + 2x_2 + x_3 \leq 430$, $3x_1 + 2x_3 \leq 460$,
 $x_1 + 4x_3 \leq 420$, $x_1, x_2, x_3 \geq 0$
 Ans : $x_1 = 100, x_2 = 125, x_3 = 80$, $Z_{max} = 950$

3. Minimize $Z = x_1 - 3x_2 + 3x_3$ subject to $3x_1 - x_2 + 2x_3 \leq 7$, $2x_1 + 4x_2 \geq -12$,
 $-4x_1 + 3x_2 + 8x_3 \leq 10$, $x_1, x_2, x_3 \geq 0$

Solution: Min $Z = \text{Max}(-Z) = -x_1 + 3x_2 - 3x_3$ subject to

$$3x_1 - x_2 + 2x_3 + S_1 = 7,$$

$$-2x_1 - 4x_2 + S_2 = 12,$$

$$-4x_1 + 3x_2 + 8x_3 + S_3 = 10,$$

$$x_1, x_2, x_3, S_1, S_2, S_3 \geq 0$$

c_j		-1	3	-3	0	0	0		
c_B	Basis	x_1	x_2	x_3	S_1	S_2	S_3	B	Min Ratio
0	S1	3	-1	2	1	0	0	7	-
0	S2	-2	-4	0	0	1	0	12	-
0	S3	4	3	8	0	0	1	10	10/3 ←
$Z_j = \sum c_B a_{ij}$		0	0 ↑	0	0	0	0	0	
$C_j = c_j - Z_j$		-1	3	-3	0	0	0	0	
0	S1	5/3	0	14/3	1	0	1/3	31/3	31/5 ←
0	S2	-22/3	0	32/3	0	1	4/3	76/3	-
3	X2	-4/3 ↑	1	8/3	0	0	1/3	10/3	-
Z_j		-4	3	8	0	0	0	10	
C_j		3	0	-11	0	0	0		
-1	X1	1	0	14/5	3/5	0	1/5	31/5	
0	S2	0	0	156/5	22/5	1	14/5	354/5	
3	X2	0	1	32/5	4/5	0	3/5	58/5	
Z_j		-1	3	82/5	9/5	0	-8/5	143/5	
C_j		0	0	-97/5	-9/5	0	0		

Since all $C_j = c_j - Z_j$ are non positive, the solution is optimal.

The optimal solution is $x_1 = 1/5$, $x_2 = 58/5$, $x_3 = 0$, Min $Z = -143/5$

4. Solve by simplex method

Maximize $Z = 2x_1 + 2x_2 + 4x_3$ subject to $2x_1 + 3x_2 + x_3 \leq 240$; $x_1 + x_2 + 3x_3 \leq 300$; $x_1 + 3x_2 + x_3 \leq 300$, $x_1, x_2, x_3 \geq 0$